

Emmett Funston

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EDUCATION

Northwestern University - B.S. in Electrical Engineering Expected June 2028
Relevant Coursework: VLSI Design, Digital Design, Mechatronics, Signals and Systems, Solid State, Circuits & Systems, Computer Systems, C/C++, Data Structures & Algorithms

SKILLS

Languages: C++, C, Python, Verilog, Assembly, Unix, Git
Digital & Analog Design: Verilog RTL, Cadence Virtuoso, Innovus, Xcelium, Genus
PCB Design & CAD: Altium Designer, KiCad, Autodesk Fusion
Simulation: LTspice, MATLAB/Simulink, Spectre
Tools: Multimeter, Oscilloscope, Power Supply, Function Generator, Logic Analyzer, Soldering, Hot Air, Crimping
Communication Protocols: CAN, I2C, SPI, UART, USB, Ethernet
Embedded: STM32, CubeIDE, Raspberry Pi Pico 2

EXPERIENCE

Schneider Electric - Engineering Intern June 2026 - September 2026

- Analyzed electrical one line diagrams, technical specifications, and application requirements for ASCO automatic transfer switch and power control systems
- Evaluated voltage/current ratings, controllers, protection features, transition modes, and accessories to configure equipment for customer power distribution applications

Bambeck Systems Inc. - Electrical Engineering Intern June 2024 - September 2024

- Calibrated QCL analyzers at Cenovus Energy and Cleveland-Cliffs IH-7 by troubleshooting electrical wiring and validating system performance, improving CO readings from 200 to 10 ppm

PROJECTS

Formula SAE Electric | Altium, C++, CAN

- Designed 4 layer STM32 CAN data acquisition PCB with multi-rail power conversion, mixed signal sensing/protection, and vehicle network integration reducing board footprint 30%
- Brought up and validated power/CAN interfaces and developed wheel sensor firmware to process ADC data, identify wheel modules, and diagnose sensor and strain gauge connectivity faults

3x3 Systolic MAC Array | Verilog, Cadence Innovus, Genus, Xcelium

- Designed 3x3 systolic MAC array in Verilog for signed 8 bit inputs and weights with signed 20 bit outputs, staged input/weight pipelines, and a 0.5 GHz target clock
- RTL in simulation, synthesis, floorplanning, placement, clock-tree synthesis, routing, and timing/physical verification in the Cadence ASIC flow yielded 0 DRC violations

4x4 SRAM Bank | Cadence Virtuoso, Spectre

- Designed a 45 nm 4x4 6T SRAM bank with differential bitlines, precharge, write, and sense amplifier circuitry
- Verified sequential read/write operation through schematic and extracted simulation, completed hierarchical DRC/LVS, and characterized frequency, energy, and layout area